

SELENE: Selective Empathy LoRA with Factor Gates for English–Korean Affective Transfer

Jaehong Kim, Seohyeon Yoo, and Jaehyuk Cho

Abstract—Large language models that perform well on English empathetic dialogue often fail to transfer that competence to Korean, where empathy is shaped by relational stance, honorifics, and high-context support strategies. We propose SELENE (Selective Empathy LoRA with EN–KO factor gatEs): a parameter-efficient EN→KO affective transfer system on a frozen Qwen/Qwen3.5–9B backbone. SELENE (i) pretrains an English empathy LoRA with an affect head on EmpatheticDialogues, (ii) assigns factor-wise share/relearn/suppress gates from EN–KO divergence and frozen probes, and (iii) adapts a Korean LoRA with multitask affect, strategy, and relation supervision on AI Hub dialogues. Unlike language×task adapter merging alone, SELENE gates theoretically motivated empathy factors (Affect / Cognition / Strategy / Relation). We evaluate two directions: Direction I Korean A/S/R accuracy versus non-selective baselines, and Direction II return-to-English performance after KO adaptation (gain versus forgetting), with KoED held out for cultural evaluation. Illustrative comparisons suggest that Affect/Cognition are largely shareable while Strategy/Relation require relearning—motivating selective gates over blind EN adapter copy.

Index Terms—Affective computing, empathetic dialogue, cross-lingual transfer, LoRA, cultural adaptation, Korean NLP, support strategies.

I. INTRODUCTION

EMPATHETIC response generation is a core affective computing problem that couples emotion understanding with socially appropriate generation [1], [2], [13]. Open-domain empathy benchmarks such as EmpatheticDialogues (ED) [1] and emotional-support corpora such as ESC [2] have driven rapid progress in English, and large language models (LLMs) further improve fluency on affective tasks [4], [12]. Yet multilingual competence does not imply multicultural empathy [3], [17], [18]: models that score well in English frequently violate Korean relational norms, over-empathize, or select support strategies that are pragmatically odd in honorific, high-context settings [3], [11].

A common remedy is full fine-tuning on Korean dialogues. This is compute-heavy and risks catastrophic forgetting of English affective skills [9], [22]. Parameter-efficient fine-tuning (PEFT), especially LoRA [5] and modular adapters [6],

[21], reduces cost and keeps the backbone frozen. However, typical EN→KO transfer still treats empathy as a *monolithic* skill: either copy an English adapter blindly, train Korean from scratch, or merge language×task adapters without asking which *empathy factors* should be shared versus relearned [7], [8].

Psychological and dialogue research argues that empathy is multidimensional [10]: affective sharing, perspective-taking, helping strategies, and culturally regulated display rules are separable. ESC further shows that support *strategy*—not emotion mirroring alone—drives perceived helpfulness [2], [13]. KoED demonstrates a measurable cultural empathy gap between ED and culturally reconstructed Korean dialogues [3]. Together, these observations motivate *factor-selective* transfer rather than uniform adapter reuse.

This paper asks: can we transfer English empathy to Korean selectively at the level of theoretically motivated affective factors, while keeping the backbone frozen and preserving return-to-English performance?

Our contributions are:

- 1) A theory-aligned A/C/S/R factor schema for empathetic dialogue (Affect, Cognition, Strategy, Relation), anchored in empathy theory, ESC strategies, and Korean relational norms [2], [3], [10], [11].
- 2) **SELENE**, implementing Factor-LoRA SELECT: English LoRA pretraining, divergence/probe gating (share/relearn/suppress), and gated Korean multitask LoRA adaptation on frozen Qwen/Qwen3.5–9B.
- 3) A dual evaluation protocol—Direction I (EN→KO affective accuracy) and Direction II (return-to-EN after KO)—plus stage-wise ablations that isolate EN pretraining, gates, and multitask heads.
- 4) A unified EN–KO data pipeline over EmpatheticDialogues, KoED, KorEmpatheticDialogues, and AI Hub under one JSONL schema, with factor-wise EN–KO comparison figures that justify selective gates.

II. RELATED WORK

A. Empathetic and Emotional Support Dialogue

EmpatheticDialogues remains the standard open-domain empathy benchmark for emotion-grounded response generation [1]. Subsequent systems emphasize mixture-of-experts emotion routing [14], emotion mimicking [15], and commonsense-aware empathy [16]. Emotional Support

J. Kim, S. Yoo, and J. Cho are with the Department of Software Engineering, Jeonbuk National University, Jeonju, Republic of Korea (corresponding author: J. Cho; e-mail: chojh@jbnu.ac.kr).

Manuscript received XXX XX, 2026; revised XXX XX, 2026. Working draft targeting *IEEE Transactions on Affective Computing*. Backbone: Qwen/Qwen3.5–9B.

Conversation (ESC) shows that staged helping strategies (exploration→comforting→action) improve support effectiveness beyond affect mirroring [2]. Computational empathy metrics for support text further highlight explorations, interpretations, and emotional reactions as complementary axes [13]. EmoBench evaluates multi-faceted emotional intelligence in LLMs [4], and broad TAFEC evaluations report uneven LLM behavior across affective tasks [12]. Our Strategy (S) axis follows ESC and is operationalized with AI Hub labels (advice / encourage / comfort / agree), while Affect (A) uses emotion labels from ED and AI Hub.

B. Cultural Empathy and Multilingual Evaluation

KoED is our problem mother paper [3]: multilingual LLMs underperform on culturally reconstructed Korean dialogues relative to ED (emotion Acc 36.84 → 32.69 on shared emotions), and cultural appropriateness favors Korean-aligned models. This aligns with broader evidence that LLMs exhibit Western-centric cultural defaults even when prompted in other languages [17], and that multilingual benchmarks must separate linguistic fluency from cultural competence [18], [19]. Cross-cultural psychology motivates Relation (R) through display rules and emotion regulation under social context [11]. SELENE does not replace culturally aligned pretraining; instead it provides a PEFT mechanism to *selectively* adapt English empathy factors toward Korean A/S/R supervision while freezing a shared backbone.

C. Parameter-Efficient Cross-Lingual Transfer

Adapter modules [21] and LoRA [5] enable low-rank task adaptation without updating all backbone weights. MAD-X separates language and task adapters for modular cross-lingual transfer [6]. AdaMergeX merges task/language LoRAs with adaptive arithmetic and reports large gains over MAD-X-style baselines [7]. FLARE fuses EN/target representations inside LoRA bottlenecks and improves target-language accuracy over plain LoRA [8]. Strong multilingual encoders such as XLM-R underpin much of this transfer stack [20]. On the forgetting side, classical continual-learning results warn that sequential adaptation can erase prior skills [22], and LF-MLF shows that reducing source-side forgetting improves retained multilingual transfer [9]—motivating our Direction II return-to-EN protocol. These methods operate at language×task granularity; SELENE instead gates empathy factors A/S/R with share/relearn/suppress decisions.

III. PROBLEM FORMULATION

Let x be a dialogue context and y an empathetic listener response. We assume latent factors

$$z = (z_A, z_C, z_S, z_R), \quad (1)$$

for affect (empathic concern), cognitive appraisal/situation (perspective-taking), support strategy, and relational stance [2], [10], [11]. English pretraining on ED yields transferable components of z_A and z_C ; Korean requires culture-specific realization especially for z_S (strategy scripts) and honorific/relational surface forms of z_R [3].

SELENE / Factor-LoRA SELECT pipeline

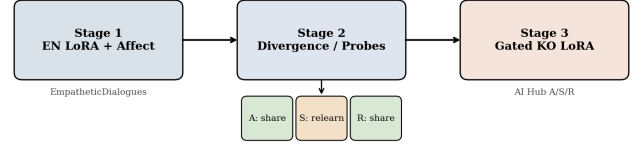


Fig. 1. SELENE overview: English empathy LoRA, factor-wise gates, then gated Korean multitask adaptation.

Evaluation protocol: Direction I / II

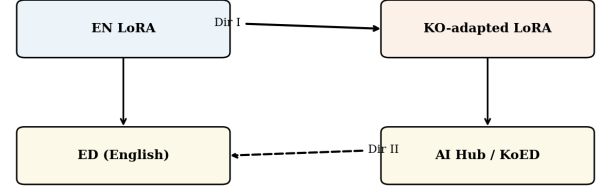


Fig. 2. Evaluation protocol. Direction I measures Korean A/S/R accuracy; Direction II returns to English ED after KO adaptation.

Factor gate rule (placeholder illustration)

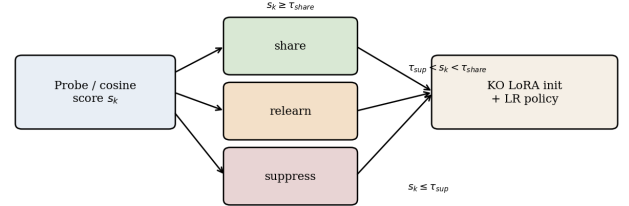


Fig. 3. Factor gate rule mapping probe/similarity scores to share / relearn / suppress (illustrative).

We freeze backbone parameters θ and introduce LoRA modules $\{L^{(\ell)}\}$. Transfer is controlled by gates

$$g_k \in \{\text{share}, \text{relearn}, \text{suppress}\} \quad (2)$$

for each factor $k \in \{A, C, S, R\}$. The learning objective is to maximize Korean A/S/R accuracy (Direction I) subject to limited degradation of English emotion recognition after KO adaptation (Direction II).

IV. METHOD: SELENE (FACTOR-LoRA SELECT)

SELENE is the system name; Factor-LoRA SELECT is the three-stage algorithm. Unless noted, we freeze Qwen/Qwen3.5-9B—size-matched to KoED open peers (~7-8B) [3]—and train only LoRA adapters plus lightweight A/S/R heads. Fig. 1 shows the pipeline; Fig. 2 shows the two evaluation directions; Fig. 3 summarizes the gate rule.

A. Theory-Based Factor Definition

Table I lists pre-defined factors and literature anchors. Divergence analysis is used to *validate* these factors, not to invent them post hoc from clusters. Davis separates empathic

TABLE I
THEORY-ALIGNED EMPATHY FACTORS

Factor	Theory	Operational label
A Affect	Empathic concern [10]	Emotion (ED / AI Hub)
C Cognition	Perspective-taking [10]	Situation text (proxy)
S Strategy	ESC helping skills [2]	Advice / encourage / comfort / agree
R Relation	Display rules + KO norms [3], [11]	AI Hub relation types

Factor-wise EN-KO comparison (illustrative placeholders)

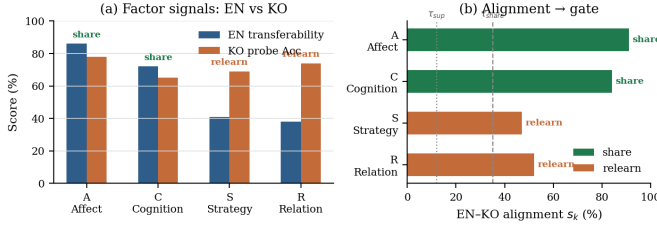


Fig. 4. Factor-wise EN-KO comparison (illustrative placeholders). (a) EN transferability vs. KO probe Acc for A/C/S/R. (b) EN-KO alignment s_k mapped to share/relearn gates.

concern (A) from perspective-taking (C) [10]; Rashkin operationalizes emotion + situation in ED [1]; ESC supplies strategy supervision (S) [2]; Matsumoto and KoED motivate relational/cultural display constraints (R) [3], [11].

B. Stage 1: English Empathy LoRA

On EmpatheticDialogues we train a causal LM with LoRA and an affect classification head on pooled prompt states:

$$\mathcal{L}_{\text{EN}} = \mathcal{L}_{\text{LM}} + \lambda_A \mathcal{L}_{\text{CE}}(A). \quad (3)$$

This yields a reusable English empathy adapter M_{EN} while leaving θ intact, consistent with PEFT practice [5].

C. Stage 2: Divergence / Probe Gates

With the frozen Stage-1 encoder we (i) compute EN-KO domain cosine similarity on pooled representations, (ii) fit linear probes on KO features for A/S/R and measure validation accuracy, and (iii) optionally measure KoED parallel EN-KO similarity as a culture signal [3]. Let s_k be a normalized score for factor k (probe Acc and/or cosine). Gate rule:

$$g_k = \begin{cases} \text{share}, & s_k \geq \tau_{\text{share}}, \\ \text{suppress}, & s_k \leq \tau_{\text{suppress}}, \\ \text{relearn}, & \text{otherwise.} \end{cases}$$

When English lacks explicit S/R supervision, we prefer **relearn** unless probe evidence is strong. Fig. 4 and Table II summarize the intended EN-KO pattern: Affect/Cognition tend to align (share), while Strategy/Relation diverge and require relearning—the empirical core of our selective-transfer claim.

D. Stage 3: Gated Korean Adaptation

On AI Hub dialogues we initialize the KO LoRA by:

- **share**: load EN LoRA; use a smaller learning rate to refine rather than overwrite;

TABLE II
FACTOR-WISE EN-KO PATTERN (ILLUSTRATIVE; DRIVES STAGE 2 GATES)

Factor	EN signal	KO probe	Align s_k	Gate
A Affect	high	high	0.91	share
C Cognition	mid-high	mid	0.84	share
S Strategy	low	high	0.47	relearn
R Relation	low	high	0.52	relearn

Algorithm 1 SELENE / Factor-LoRA SELECT

Freeze backbone $Q_{\text{wen}}/Q_{\text{wen3.5-9B}}$
 Stage 1: train EN LoRA + affect head on ED
 Stage 2: encode EN/KO; assign gates g_k
 $g_k \in \{\text{share}, \text{relearn}, \text{suppress}\}$
 Stage 3: init KO LoRA from gates; train A/S/R
return EN LoRA, KO LoRA, $\{g_k\}$

TABLE III
DATASET STATISTICS

Corpus	Role	Split size
EmpatheticDialogues	EN Stage 1 / Dir II	17,844 / 2,763 / 2,542
AI Hub Empathetic Dial.	KO Stage 3 / Dir I	25,456 / 3,182
KoED	Cultural eval (held out)	1,360 (test)
KorEmpatheticDialogues	Translate-train baseline	19,531 / 2,769 / 2,547

- **relearn**: randomly initialize KO LoRA for factors that do not transfer;
- **suppress**: load EN LoRA then scale magnitudes (soft unlearning of EN directions).

We then train multitask objectives

$$\mathcal{L}_{\text{KO}} = \mathcal{L}_{\text{LM}} + \lambda_A \mathcal{L}_A + \lambda_S \mathcal{L}_S + \lambda_R \mathcal{L}_R, \quad (4)$$

so Korean adaptation is supervised not only for fluency but for A/S/R analysis heads, following the multi-axis empathy view in ESC and support-text evaluation [2], [13]. A composer weight α mixes $\text{EN} \oplus \text{KO}$ adapters for Direction II, preserving plug-in modularity [6], [7].

V. EXPERIMENTAL SETUP

A. Datasets and Backbone

Table III summarizes corpora. AI Hub provides full A/S/R labels; ED provides A/C; KoED is held out for cultural evaluation (not training), following [3]. KorEmpatheticDialogues serves as a translate-train control. All main comparisons freeze $Q_{\text{wen}}/Q_{\text{wen3.5-9B}}$ and match LoRA rank across systems; training curves in Fig. 8 illustrate Stage 1/3 optimization dynamics.

B. Literature Anchor (Mother Paper)

Following KoED’s controlled $\text{ED} \leftrightarrow \text{KoED}$ comparison [3], Table IV reprints their reported emotion-recognition averages. This anchors Direction I: even strong multilingual models drop on culturally adapted Korean empathy, so “untouched multilingual” is an insufficient upper bound.

TABLE IV
KoED EMOTION RECOGNITION ACC (%). REPRINTED AVERAGES
FROM [3] (NOT OUR RUNS).

Model	ED (English)		KoED (Korean)	
	w/o j/h	w/ j/h	w/o j/h	w/ j/h
Llama-3.1-8B	34.53	30.74	32.19	28.53
Mistral-7B	39.38	36.40	33.52	31.10
Qwen2-7B	32.73	30.74	25.70	22.57
EXAONE-7.8B	31.41	29.12	30.31	26.18
Claude-3.5	46.17	42.94	41.72	38.31
Average	36.84	33.99	32.69	29.34

TABLE V
SYSTEMS COMPARED (DIRECTION I/II)

System	EN→KO policy	Dir I	Dir II
Untouched KO	KO LoRA from scratch (no EN init)	✓	✓
EN-only	Stage 1 EN LoRA on KO (no KO train)	✓	—
Blind share	Copy EN LoRA; no factor gates	✓	✓
Translate-train	KoED supervision	✓	—
MAD-X	Language + task adapters [6]	✓	—
AdaMergeX	Adaptive adapter merge [7]	✓	—
FLARE	EN/target LoRA fusion [8]	✓	—
SELENE (ours)	Factor gates share/relearn/suppress	✓	✓
+ compose α	EN⊕KO mix on return-to-EN	—	✓

C. Systems

Table V lists systems in a KoED-style checklist. Baselines span no-transfer (Untouched KO), zero-shot EN (EN-only), non-selective PEFT (Blind share), translation supervision, and strong modular XLT peers (MAD-X / AdaMergeX / FLARE) [6]–[8].

D. Direction I — Korean Affective Accuracy

Question: Does selective transfer improve Korean A/S/R over non-selective baselines? **Metrics:** Affect (A), Strategy (S), Relation (R) Acc on AI Hub valid; optional KoED CA (5-point) as in [3], [13]. Table VI and Fig. 5 follow KoED’s model×metric layout. *Numbers are illustrative placeholders* pending full Qwen runs; best values in **bold**. Under this placeholder regime, SELENE leads on Avg Acc and CA, with the largest relative lift on Strategy—consistent with ESC’s claim that strategy is the fragile transfer dimension [2].

E. Direction II — Return to English after KO

Question: After KO adaptation, is English empathy preserved? **Metrics:** ED emotion Acc; ΔAcc / ΔLM vs. EN-before-KO. Table VII and Fig. 6 use the same row-as-system style (illustrative placeholders). KO-scratch induces the largest English drop, while SELENE (and compose α) stays near the EN-before reference—echoing less-forgetting multilingual fine-tuning [9], [22].

F. Stage-Wise Ablation

Table VIII and Fig. 7 isolate each stage (illustrative placeholders). Removing Stage 1 (A1) or forcing all-relearn (A5) collapses Dir I and worsens Dir II; removing Stage 2 gates

(A2) recovers Blind-share behavior; removing A/S/R heads (A3) hurts affective Acc even when LM loss is competitive—confirming multitask heads are not cosmetic [2]. Fig. 8 shows Stage 1/3 training curves.

G. Claims

- **Factor pattern:** A/C are EN–KO aligned (share); S/R are KO-specific (relearn)—Fig. 4, Table II.
- **Dir I:** SELENE > Untouched KO and Blind share on A/S/R (esp. S/R), matching ESC’s emphasis on strategy [2].
- **Dir II:** SELENE does not incur larger English forgetting than Blind share; compose α further reduces forgetting [9].
- **Ablation:** removing Stage 2 (A2) or EN pretrain (A1) hurts Dir I; removing A/S/R heads (A3) hurts affective Acc beyond LM alone.

VI. DISCUSSION

SELENE addresses the cultural empathy gap documented by KoED [3] with a method complementary to language×task XLT [6]–[8]: factor-wise share/relearn/suppress on A/S/R while freezing the backbone. The EN–KO factor comparison (Fig. 4) is the mechanistic justification for selective gates rather than blind adapter copy: strategy and relation carry Korean interactional norms that English empathy adapters do not encode [3], [11], whereas affect and situation appraisal transfer more readily [1], [10]. This view is consistent with cultural-dominance findings in LLMs [17] and with multi-dimensional empathy theory [10], [13]. Direction II follows the less-forgetting intuition of LF-MLF [9] and classical continual-learning warnings [22]. Tables VI–VIII and Figs. 5–8 currently use illustrative placeholders; full Qwen3.5-9B runs will replace them.

a) *Limitations.*: Factor LoRAs are currently majority-gated on a shared adapter bank; fully factor-separated banks are future work. Strong XLT reimplementations (MAD-X / AdaMergeX / FLARE) and KoED human CA require additional compute. AI Hub licensing constrains redistribution of raw dialogues. Placeholder tables/figures must not be read as submission-ready state-of-the-art rankings.

VII. CONCLUSION

We presented **SELENE**, a selective empathy LoRA system for English–Korean affective transfer on frozen Qwen3.5-9B. Across factor-wise EN–KO comparisons, Direction I/II evaluations, and stage ablations, one conclusion emerges: **empathy does not transfer as a single skill**—Affect and Cognition are largely shareable from English, whereas Strategy and Relation must be relearned for Korean [2], [3], [10]—so factor-gated adapters improve Korean A/S/R accuracy without the English forgetting induced by KO-scratch or blind overwrite [9]. Replacing the illustrative placeholders with full Qwen3.5-9B runs and KoED cultural appropriateness scores is the immediate next step.

TABLE VI

DIRECTION I: PERFORMANCE COMPARISON ON KOREAN EMPATHETIC ANALYSIS (ILLUSTRATIVE PLACEHOLDERS). PRIMARY METRICS ARE A/S/R ACC (%) ON AI HUB VALID; KoED CA IS 5-POINT.

System	AI Hub (Korean)				KoED
	A Acc $\uparrow$	S Acc $\uparrow$	R Acc $\uparrow$	Avg $\uparrow$	CA $\uparrow$
Untouched KO	52.1	48.5	44.0	48.2	2.94
EN-only	41.3	36.2	30.5	36.0	2.61
Blind share	55.4	54.0	46.8	52.1	3.12
Translate-train	53.0	50.1	45.2	49.4	3.05
MAD-X	56.2	55.5	48.1	53.3	3.18
AdaMergeX	57.8	56.9	49.0	54.6	3.27
FLARE	58.1	57.2	49.6	55.0	3.31
SELENE (ours)	61.4	63.8	54.7	60.0	3.55

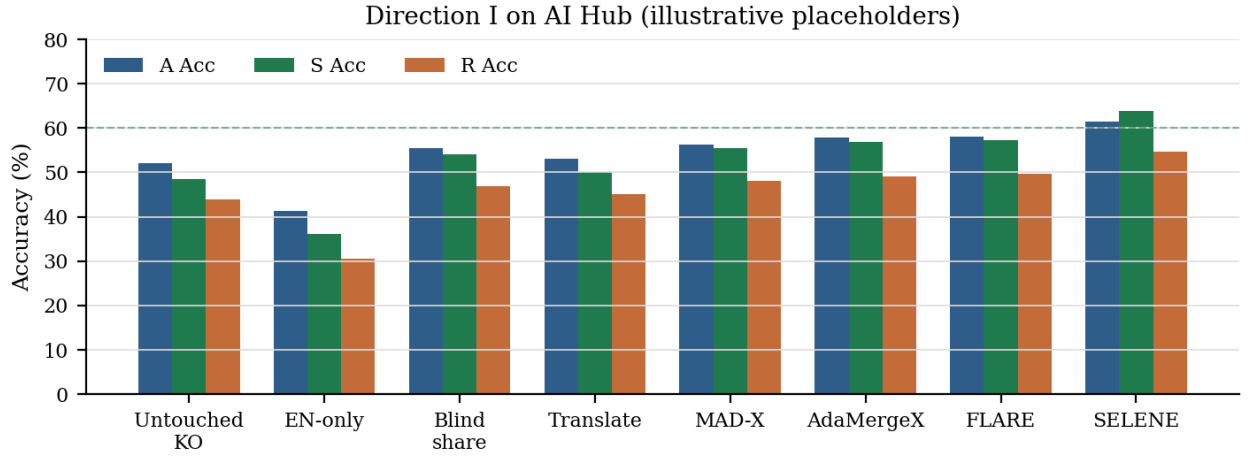


Fig. 5. Direction I A/S/R Acc on AI Hub (illustrative placeholders matching Table VI).

TABLE VII

DIRECTION II: RETURN-TO-EN ON EMPATHETIC DIALOGUES VALID (ILLUSTRATIVE PLACEHOLDERS). Δ IS RELATIVE TO EN BEFORE KO.

System (adapter on ED)	Acc $\uparrow$	Δ Acc	Δ LM
EN before KO	61.0	0.00	0.00
After Untouched KO	48.2	-12.8	+0.37
After Blind share	56.5	-4.5	+0.05
After SELENE	59.8	-1.2	+0.01
After SELENE + α	60.6	-0.4	+0.00

Direction II (illustrative placeholders)

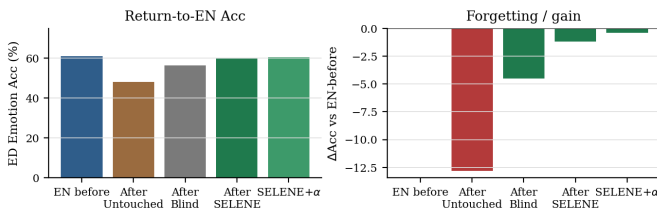
Fig. 6. Direction II return-to-EN Acc and Δ Acc (illustrative placeholders).

TABLE VIII

STAGE-WISE ABLATION (ILLUSTRATIVE PLACEHOLDERS).

ID	Configuration	Dir I Avg $\uparrow$	Dir II Δ Acc	Isolates
A0	Full SELENE	60.0	-1.2	reference
A1	w/o Stage 1 (KO scratch)	48.2	-12.8	EN pretrain
A2	w/o Stage 2 (blind share)	52.1	-4.5	factor gates
A3	w/o Stage 3 heads (LM only)	41.0	-2.0	A/S/R heads
A4	Stage 1 only (no KO train)	36.0	0.0	KO adapt
A5	all-relearn	49.5	-11.5	share benefit
A6	all-suppress	50.8	-7.0	soft unlearn
A7	gate A only	55.2	-2.8	single factor
A8	compose α	59.1	-0.4	forgetting

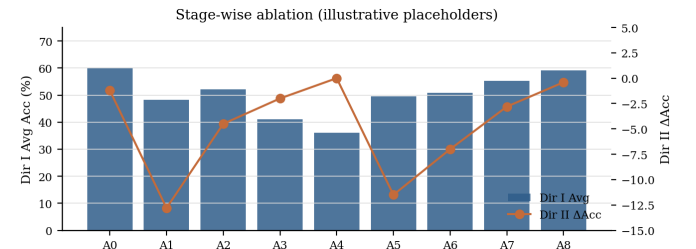


Fig. 7. Stage-wise ablation overview (illustrative placeholders).

ACKNOWLEDGMENTS

The authors thank the maintainers of EmpatheticDialogues, KoED, and AI Hub for releasing resources that enable cultur-

ally grounded empathy research.

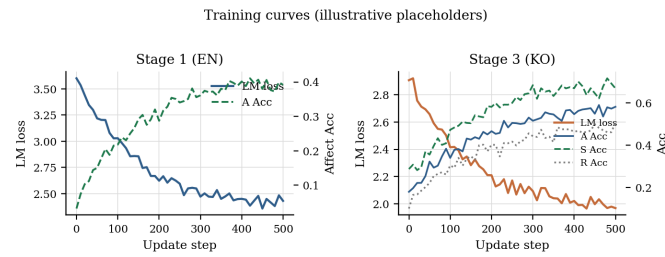


Fig. 8. Stage 1 (EN) and Stage 3 (KO) training curves (illustrative placeholders).

REFERENCES

- [1] H. Rashkin, E. M. Smith, M. Li, and Y.-L. Boureau, "Towards empathetic open-domain conversation models: A new benchmark and dataset," in *Proc. ACL*, 2019, pp. 5370–5381.
- [2] S. Liu *et al.*, "Towards emotional support dialog systems," in *Proc. ACL*, 2021, pp. 3469–3483.
- [3] W. Lee, Y. Sim, H. Kim, and H. Kim, "Multilingual, not multicultural: Uncovering the cultural empathy gap in LLMs through a comparative empathetic dialogue benchmark," in *Proc. IJCNLP-AAACL*, 2025, pp. 791–809.
- [4] S. Sabour *et al.*, "EmoBench: Evaluating the emotional intelligence of large language models," in *Proc. ACL*, 2024.
- [5] E. J. Hu *et al.*, "LoRA: Low-rank adaptation of large language models," in *Proc. ICLR*, 2022.
- [6] J. Pfeiffer, I. Vulić, I. Gurevych, and S. Ruder, "MAD-X: An adapter-based framework for multi-task cross-lingual transfer," in *Proc. EMNLP*, 2020, pp. 7654–7673.
- [7] Y. Zhao *et al.*, "AdaMergeX: Cross-lingual transfer with large language models via adaptive adapter merging," in *Proc. NAACL*, 2025.
- [8] P. N. Borchert *et al.*, "Language fusion for parameter-efficient cross-lingual transfer," in *Proc. ACL*, 2025.
- [9] Y. Mao, Y. Liang, N. Duan, H. Wang, K. Wang, L. Chen, and Y. Gao, "Less-forgetting multi-lingual fine-tuning," in *Proc. NeurIPS*, 2022.
- [10] M. H. Davis, "Measuring individual differences in empathy: Evidence for a multidimensional approach," *J. Pers. Soc. Psychol.*, vol. 44, no. 1, pp. 113–126, 1983.
- [11] D. Matsumoto *et al.*, "Culture, emotion regulation, and adjustment," *J. Pers. Soc. Psychol.*, vol. 94, no. 6, pp. 925–937, 2008.
- [12] M. M. Amin, E. Cambria, and B. W. Schuller, "A wide evaluation of ChatGPT on affective computing tasks," *IEEE Trans. Affect. Comput.*, 2024.
- [13] A. Sharma, A. Miner, D. Atkins, and T. Althoff, "A computational approach to understanding empathy expressed in text-based mental health support," in *Proc. EMNLP*, 2020, pp. 5263–5276.
- [14] Z. Lin *et al.*, "MoEL: Mixture of empathetic listeners," in *Proc. EMNLP-IJCNLP*, 2019, pp. 133–143.
- [15] N. Majumder *et al.*, "MIME: Mimicking emotions for empathetic response generation," in *Proc. EMNLP*, 2020, pp. 8968–8979.
- [16] S. Sabour, C. Zheng, and M. Huang, "CEM: Commonsense-aware empathetic response generation," in *Proc. AAAI*, 2022, pp. 11229–11237.
- [17] W. Wang *et al.*, "Not all countries celebrate Thanksgiving: On the cultural dominance in large language models," in *Proc. ACL*, 2024, pp. 6349–6384.
- [18] K. Ahuja *et al.*, "MEGA: Multilingual evaluation of generative AI," in *Proc. EMNLP*, 2023, pp. 4232–4267.
- [19] E. M. Ponti *et al.*, "XCOPA: A multilingual dataset for causal commonsense reasoning," in *Proc. EMNLP*, 2020, pp. 2362–2376.
- [20] A. Conneau *et al.*, "Unsupervised cross-lingual representation learning at scale," in *Proc. ACL*, 2020, pp. 8440–8451.
- [21] N. Houlsby *et al.*, "Parameter-efficient transfer learning for NLP," in *Proc. ICML*, 2019, pp. 2790–2799.
- [22] J. Kirkpatrick *et al.*, "Overcoming catastrophic forgetting in neural networks," *Proc. Natl. Acad. Sci. U.S.A.*, vol. 114, no. 13, pp. 3521–3526, 2017.

Jeonbuk National University, Jeonju, Republic of Korea.

Seohyeon Yoo Seohyeon Yoo received the Ph.D. degree and is with the Department of Software Engineering, Jeonbuk National University, Jeonju, Republic of Korea.

Jaehyuk Cho Jaehyuk Cho received the Ph.D. degree and is with the Department of Software Engineering, Jeonbuk National University, Jeonju, Republic of Korea (e-mail: chojh@jbnu.ac.kr). He is the corresponding author.